WEB-BASED ML MODEL EVALUATOR

TEST PLAN DOCUMENT - Updated for Sprint 3 (BL-12)

1. Introduction

This document replaces the earlier test plan, which covered only UC-01 to UC-05 and described the first version of the system. It now covers all thirteen use cases (UC-01 to UC-13), the security requirements and the non-functional requirements of the SRS, so that every requirement FR-01 to FR-08 and NFR-01 to NFR-05 has at least one test case.

What changed in the system since the earlier plan: registration and login with Analyst and Admin roles, an Admin area (suspend, reactivate, change role, audit log), per-user data isolation, SQLite storage, charts (confusion matrix, ROC, residual, predicted-vs-actual), saved evaluation history, comparison, PDF reports, and compatibility results shown when a file is validated.

2. Requirement Traceability

Every test case lists the SRS requirement it verifies. The table below is the reverse view: for each requirement, the test cases that cover it.

Requirement	Meaning	Covered by test cases	Count
FR-01	Register and log in; roles	SEC_01, SEC_02, SEC_03, SEC_04, SEC_06, 01_UT_01, 01_UT_02, 01_UT_03, 01_UT_04, 01_ST_01, 02_UT_01, 02_UT_02, 02_IT_01, 02_IT_02, 02_ST_01, 13_ST_01	16
FR-02	Upload and validate dataset and models; compatibility	03_UT_01, 03_UT_02, 03_UT_03, 03_UT_04, 03_IT_01, 03_IT_02, 04_UT_01, 04_UT_02, 04_UT_03, 04_IT_01, 04_IT_02, 04_IT_03, 06_IT_02, 06_ST_01	14
FR-03	Select task type; task/model/dataset check	05_UT_01, 05_IT_01, 05_ST_01, 06_IT_02, 06_ST_01	5
FR-04	Run model and compute metrics	06_UT_01, 06_UT_02, 06_UT_03, 06_ST_01, 06_ST_02, 07_UT_01	6
FR-05	Visualizations	06_UT_03, 06_ST_01, 08_UT_01, 08_UT_02, 08_IT_01, 08_ST_01, 08_ST_02	7
FR-06	Compare evaluations	09_UT_01, 09_IT_01, 09_IT_02, 09_ST_01	4
FR-07	Saved evaluations, history and reports	06_IT_01, 07_IT_01, 10_UT_01, 10_UT_02, 11_UT_01, 11_UT_02, 11_UT_03, 11_IT_01, 11_ST_01	9
FR-08	Admin: accounts, roles, audit log	SEC_06, SEC_07, 02_IT_02, 02_IT_03, 02_IT_04, 12_UT_01, 12_UT_02, 12_IT_01, 12_IT_02, 12_ST_01, 13_UT_01, 13_IT_01, 13_IT_02, 13_IT_03, 13_ST_01	15
NFR-01	Performance (10 s at 50,000 rows / 100 MB)	03_UT_04, NF_PF_01, NF_PF_02, NF_PF_03	4
NFR-02	Security: data isolation between accounts	SEC_08, SEC_09, SEC_10, SEC_11, SEC_12	5
NFR-03	Reliability: clear errors	03_UT_02, 04_UT_02, 04_UT_03, 04_IT_01, 06_ST_03, 07_UT_02, 11_ST_01, 12_ST_01, NF_ER_01, NF_ER_02, NF_ER_03, NF_ER_04	12
NFR-04	Usability	NF_US_01, NF_US_02	2
NFR-05	Maintainability / extensibility	NF_EX_01	1
Security	Security (CSRF and access control)	SEC_05	1

Note: the acceptance criteria of FR-02 to FR-04 require compatibility to be checked before an evaluation; this is covered by 04_IT_02, 04_IT_03, 05_IT_01, 05_ST_01 and 06_IT_02.

3. Test Approach

Levels. The Test Case ID shows the level: **UT** = unit (one function or service on its own), **IT** = integration (several modules or the API together), **ST** = system (a full user workflow). The prefix is the use case number (01 = UC-01 Register, and so on). Cases that are not tied to one use case use **SEC** (security), **NF_PF** (performance), **NF_ER** (error handling), **NF_US** (usability) and **NF_EX** (extensibility).

Automation. Most cases are automated with pytest in `backend/tests`; the column 'Automated test' gives the exact test function. The suite contains additional supporting tests (for example hashing, e-mail case handling, duplicate-row counting) that are not listed here; they run together with the listed ones. Every test runs against its own temporary database and folders, so results do not depend on each other. 2 cases are manual (NF_PF_03 and NF_US_01) because they need a very large model file and a person who has not seen the app.

How to run. `cd backend; pip install -r requirements-dev.txt; python -m pytest tests -v`. For coverage (BL-15): `python -m pytest tests --cov=. --cov-branch --cov-report=term-missing --cov-report=html`.

4. Test Environment and Pass / Fail Criteria

Environment. Python 3.13 (the team may also use a newer version), Flask, pandas, scikit-learn, joblib, matplotlib, reportlab, SQLite. Sample files are in `backend/sample_data`.

Pass. The actual result equals the expected result. For NFR-01 the evaluation finishes in under 10 seconds for up to 50,000 rows with one user (the SRS asks for 95 % of runs).

Fail. Any difference from the expected result. Each failure is logged as a defect with the test case ID and fixed under BL-13; the case is then run again.

Status of this document. The plan was executed on 4 November 2026 (BL-13). The results are in the columns 'Actual Result' and 'Test Result' and are summarised in section 7.

5. Test Cases

5.1 Security Validation

Authentication, role-based access, CSRF protection and isolation of each account's data (SRS FR-01, FR-08, NFR-02).

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
SEC_01	FR-01	Passwords are stored hashed, never as plain text	App running with empty database.	Register an account; read the stored user record.	Password Str0ngPass!	Stored value is a hash and differs from the typed password.	`test_auth::test_passwords_are_stored_hashed`	As expected. 1 automated test(s) passed.	**Pass**
SEC_02	FR-01	Registration can never create an Admin	Registration page open.	Submit the registration form with role=admin added to the request.	Valid details plus role=admin	Account is created as Analyst only.	`test_auth::test_01_IT_02_registration_cannot_create_an_admin`	As expected. 1 automated test(s) passed.	**Pass**
SEC_03	FR-01	Anonymous visitor cannot reach workspace pages or the API	Not logged in.	Open /dashboard, /models and /api/models without logging in.	None	Pages redirect to login; the API returns 401 JSON.	`test_auth::test_pages_redirect_anonymous_visitors_to_login` `test_auth::test_api_returns_401_json_for_anonymous_visitors`	As expected. 13 automated test(s) passed.	**Pass**
SEC_04	FR-01	Login cannot redirect to another website	Login page open.	Log in with next=https://other-site.example.	External next URL	User lands on an internal page, not the external site.	`test_auth::test_login_does_not_redirect_to_other_sites`	As expected. 3 automated test(s) passed.	**Pass**
SEC_05	Security	Request without a valid CSRF token is rejected	Logged in as Analyst.	Send a POST with no token, then with a wrong token; try the login form without a token.	Missing / wrong token	Each request is rejected and nothing changes.	`test_auth::test_post_without_csrf_token_is_rejected` `test_auth::test_post_with_wrong_csrf_token_is_rejected` `test_auth::test_login_form_without_csrf_token_is_rejected`	As expected. 3 automated test(s) passed.	**Pass**
SEC_06	FR-01 / FR-08	Analyst cannot open Admin pages or call the Admin API	Logged in as Analyst.	Open /admin/users and /admin/audit; call the admin API endpoints.	Analyst session	Access is refused; no data is returned.	`test_admin::test_analyst_cannot_use_admin_api` `test_admin::test_analyst_cannot_open_admin_pages`	As expected. 4 automated test(s) passed.	**Pass**
SEC_07	FR-08	Analyst cannot change roles or make themselves Admin	Logged in as Analyst.	Call the role and status endpoints for own and other accounts; try creating an account.	Analyst session	All attempts are refused; roles are unchanged.	`test_admin::test_analyst_cannot_change_roles_or_statuses_by_calling_the_api` `test_admin::test_analyst_cannot_make_themselves_admin` `test_admin::test_analyst_cannot_create_accounts`	As expected. 3 automated test(s) passed.	**Pass**
SEC_08	NFR-02	Analyst B cannot see Analyst A's files in any list	A has uploaded a model, dataset and evaluation; B is logged in.	As B, open the model, dataset and evaluation lists.	Two analyst accounts	B's lists show only B's own records.	`test_isolation::test_lists_only_show_your_own_records`	As expected. 1 automated test(s) passed.	**Pass**
SEC_09	NFR-02	Guessing another user's ID returns 404 (validate, delete, evaluate)	A owns model, dataset, evaluation; B is logged in.	As B, call validate, delete and evaluate using A's IDs.	A's record IDs	404 Not Found; A's data is untouched.	`test_isolation::test_bob_cannot_validate_alices_model_or_dataset_by_id` `test_isolation::test_bob_cannot_delete_alices_files` `test_isolation::test_bob_cannot_evaluate_with_alices_files`	As expected. 3 automated test(s) passed.	**Pass**
SEC_10	NFR-02	Guessing another user's evaluation, chart, comparison or report URL returns 404	A owns a completed evaluation; B is logged in.	As B, open A's evaluation page, chart, comparison and report URLs; include A's ID in B's own report request.	A's evaluation ID	404 Not Found; no part of A's data is included.	`test_isolation::test_bob_cannot_open_alices_evaluation_by_id` `test_isolation::test_bob_cannot_fetch_alices_charts_comparison_or_report` `test_isolation::test_a_report_cannot_smuggle_in_another_users_evaluation`	As expected. 3 automated test(s) passed.	**Pass**
SEC_11	NFR-02	Same file name used by two users does not clash or leak	A and B each upload a file with the same name.	List files as A and as B.	Same file name	Each user sees only their own file.	`test_isolation::test_uploaded_file_names_are_scoped_to_the_owner_in_listings`	As expected. 1 automated test(s) passed.	**Pass**

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
SEC_12	NFR-02	Admin cannot read an Analyst's data; compatibility results never mention other users' files	Analyst has uploaded data; Admin is logged in.	As Admin, request the Analyst's model, dataset and evaluation by ID; validate a file as another user.	Analyst record IDs	Not found; no other user's file names appear anywhere.	`test_isolation::test_admin_cannot_read_an_analysts_data` `test_isolation::test_cross_user_dataset_compatibility_report_never_mentions_other_users_files`	As expected. 2 automated test(s) passed.	**Pass**

5.2 Functional Test Cases by Use Case

UC-01 Register

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
01_UT_01	FR-01	Blank required fields are rejected	Registration page open.	Leave required fields blank; submit.	Empty name / email / password	Each missing field is named; no account is created.	`test_auth::test_01_UT_01_blank_required_fields_are_rejected`	As expected. 1 automated test(s) passed.	**Pass**
01_UT_02	FR-01	Invalid email and short password are rejected	Registration page open.	Submit once with a bad email, once with a password under 8 characters.	abc / name@; password 'short'	Clear message for each; no account is created.	`test_auth::test_01_UT_02_invalid_email_is_rejected` `test_auth::test_01_UT_02_short_password_is_rejected`	As expected. 5 automated test(s) passed.	**Pass**
01_UT_03	FR-01	Duplicate email is rejected (ignoring letter case)	An account exists for a@x.com.	Register again with A@X.com.	A@X.com	'Account already exists' message; no second account.	`test_auth::test_01_UT_03_duplicate_email_is_rejected_ignoring_case`	As expected. 1 automated test(s) passed.	**Pass**
01_UT_04	FR-01	Mismatched password confirmation is rejected	Registration page open.	Enter two different passwords; submit.	Passwords that differ	Rejected with a clear message.	`test_auth::test_mismatched_password_confirmation_is_rejected`	As expected. 1 automated test(s) passed.	**Pass**
01_ST_01	FR-01	Valid registration creates an Analyst and signs in	Registration page open.	Fill valid details; submit.	Valid name, email, 8+ char password	Analyst account created and user lands on the dashboard.	`test_auth::test_01_IT_03_valid_registration_creates_an_analyst_and_signs_in`	As expected. 1 automated test(s) passed.	**Pass**

UC-02 Login

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
02_UT_01	FR-01	Blank login fields are rejected	Login page open.	Submit with empty email or password.	Empty fields	Message asks for email and password.	`test_auth::test_02_UT_01_blank_login_fields_are_rejected`	As expected. 1 automated test(s) passed.	**Pass**
02_UT_02	FR-01	Wrong password and unknown email give the same message	One Analyst account exists.	Log in with wrong password; log in with unknown email.	Wrong password / unknown email	Same generic message (does not reveal which part was wrong).	`test_auth::test_02_UT_03_wrong_password_and_unknown_email_give_the_same_message`	As expected. 1 automated test(s) passed.	**Pass**
02_IT_01	FR-01	Analyst login opens the Analyst dashboard	Analyst account exists.	Log in as Analyst.	Valid Analyst credentials	Analyst dashboard opens.	`test_auth::test_02_IT_01_analyst_login_goes_to_the_analyst_dashboard`	As expected. 1 automated test(s) passed.	**Pass**
02_IT_02	FR-01 / FR-08	Admin login opens the dashboard with Admin access	Admin account exists.	Log in as Admin.	Valid Admin credentials	Admin is signed in and can open Admin pages.	`test_auth::test_02_IT_02_admin_login_goes_to_the_admin_dashboard`	As expected. 1 automated test(s) passed.	**Pass**
02_IT_03	FR-08	Suspended account cannot log in; works again once reactivated	Account suspended by Admin.	Try to log in; reactivate; log in again.	Suspended account	Login refused while suspended; succeeds after reactivation.	`test_auth::test_02_IT_03_suspended_account_cannot_log_in` `test_auth::test_reactivated_account_can_log_in_again`	As expected. 2 automated test(s) passed.	**Pass**
02_IT_04	FR-08	Suspending an account ends its open session at once	User is logged in; Admin suspends the account.	User clicks any page after suspension.	Open session	User is sent out immediately (no wait for session expiry).	`test_auth::test_suspending_an_account_ends_its_open_session_immediately`	As expected. 1 automated test(s) passed.	**Pass**
02_ST_01	FR-01	Logout ends the session; login returns to requested page	Logged in.	Log out, then open a workspace page; log in again.	None	Logout blocks access; after login the user returns to the page asked for.	`test_auth::test_logout_ends_the_session` `test_auth::test_login_redirects_back_to_the_requested_page`	As expected. 2 automated test(s) passed.	**Pass**

UC-03 Upload Dataset

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
03_UT_01	FR-02	CSV accepted, other file types rejected	Logged in as Analyst.	Upload a .csv; then upload a .txt.	sample_classification.csv; notes.txt	CSV accepted; other type rejected naming the accepted type.	`test_dataset_service::test_03_UT_01_csv_dataset_is_accepted` `test_dataset_service::test_03_UT_01_unsupported_dataset_formats_are_rejected`	As expected. 5 automated test(s) passed.	**Pass**
03_UT_02	FR-02 / NFR-03	Corrupt dataset is rejected with a reason	Logged in.	Upload a .csv that is not valid CSV.	Corrupt file	Rejected with a specific reason; nothing stored.	`test_dataset_service::test_03_UT_02_corrupt_dataset_is_rejected` `test_api_workflow::test_03_IT_02_corrupt_dataset_is_rejected_with_a_reason`	As expected. 4 automated test(s) passed.	**Pass**
03_UT_03	FR-02	Target column must be given and must exist	Logged in.	Upload without a target; upload with a target name not in the file.	Blank target; wrong target	Rejected; message lists the real column names.	`test_dataset_service::test_03_UT_03_target_column_must_be_specified` `test_dataset_service::test_03_UT_03_target_column_must_exist_and_error_lists_columns` `test_api_workflow::test_03_ST_02_dataset_upload_without_a_target_is_rejected`	As expected. 3 automated test(s) passed.	**Pass**
03_UT_04	FR-02 / NFR-01	Row limit: 50,000 accepted, 50,001 rejected	Logged in.	Upload a file with exactly 50,000 rows; then one with 50,001.	50,000 and 50,001 rows	First accepted; second rejected with the limit named.	`test_dataset_service::test_dataset_exactly_at_row_limit_is_accepted` `test_dataset_service::test_dataset_over_row_limit_is_rejected`	As expected. 2 automated test(s) passed.	**Pass**
03_IT_01	FR-02	Dataset with missing target values is marked invalid	Dataset uploaded.	Validate a dataset where some target cells are empty.	bad_dataset_missing_target.csv	Marked invalid with the reason; cannot be used for evaluation.	`test_dataset_service::test_missing_target_values_make_dataset_invalid`	As expected. 1 automated test(s) passed.	**Pass**
03_IT_02	FR-02	Valid dataset passes validation with a schema summary	Dataset uploaded.	Validate a clean dataset.	sample_classification.csv	Marked valid; row count and columns shown; missing feature values are reported but allowed.	`test_dataset_service::test_clean_dataset_passes_validation_with_schema_summary` `test_dataset_service::test_missing_values_in_features_are_reported_but_still_valid`	As expected. 2 automated test(s) passed.	**Pass**

UC-04 Upload Model

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
04_UT_01	FR-02	Only .pkl and .joblib models are accepted	Logged in.	Upload a .pkl; upload a .txt.	sample_classifier .pkl; notes.txt	Supported format accepted; others rejected.	`test_model_service::test_04_UT_01_supported_model_formats_are_accepted` `test_model_service::test_04_UT_01_unsupported_model_formats_are_rejected`	As expected. 5 automated test(s) passed.	**Pass**
04_UT_02	FR-02 / NFR-03	No file or empty file is rejected	Logged in.	Submit with no file; submit an empty file.	No file / 0-byte file	Rejected with a clear message.	`test_model_service::test_04_UT_02_no_file_selected_is_rejected` `test_model_service::test_empty_model_file_is_rejected`	As expected. 2 automated test(s) passed.	**Pass**

04_UT_03	FR-02 / NFR-03	Size limit: 100 MB accepted, above 100 MB rejected	Logged in.	Upload a file of exactly 100 MB; then one slightly larger.	100 MB and 100 MB + 1 byte	First accepted; second rejected with the limit named (HTTP 413 as JSON).	`test_model_service::test_model_exactly_at_size_limit_is_accepted`
							`test_model_service::test_model_over_size_limit_is_rejected`
							`test_api_workflow::test_oversized_upload_returns_413_json`
								As expected. 3 automated test(s) passed.	**Pass**

04_IT_01	FR-02 / NFR-03	Corrupt model is flagged invalid with a reason	Corrupt .pkl uploaded.	Validate the model.	Corrupt model file	Marked invalid with a specific reason.	`test_model_service::test_corrupt_model_file_is_marked_invalid`
							`test_model_service::test_loadable_object_without_predict_is_marked_invalid`
							`test_api_workflow::test_04_IT_03_invalid_model_file_is_rejected_immediately_with_a_reason`
								As expected. 3 automated test(s) passed.	**Pass**

04_IT_02	FR-02	Model compatible with dataset passes; missing features fail	Valid dataset uploaded.	Check a matching model; check a model that needs a column the dataset lacks.	Matching / non-matching model	Compatible model accepted; the other rejected naming the missing columns.	`test_evaluation_service::test_04_IT_02_compatible_model_is_accepted`
							`test_evaluation_service::test_04_IT_02_model_with_features_missing_from_dataset_is_rejected`
							`test_evaluation_service::test_04_ST_02_model_with_wrong_feature_count_is_rejected`
								As expected. 3 automated test(s) passed.	**Pass**

04_IT_03	FR-02	Compatibility with each existing file is shown at validation	Model(s) and dataset(s) uploaded.	Validate a model, then a dataset.	One model, two datasets	Result lists fit with each of the user's existing datasets (or models).	`test_compatibility_at_upload::test_validated_model_reports_fit_with_each_existing_dataset` `test_compatibility_at_upload::test_validated_dataset_reports_fit_with_each_existing_model`	As expected. 2 automated test(s) passed.	**Pass**

UC-05 Select Task Type

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result

05_UT_01	FR-03	Classification and regression accepted; unknown task rejected	Valid model and dataset.	Select each task; submit an unknown task name.	classification / regression / 'cluster'	First two accepted; unknown rejected.	`test_evaluation_service::test_05_UT_02_classification_task_is_accepted`
							`test_evaluation_service::test_05_UT_03_regression_task_is_accepted`
							`test_evaluation_service::test_unknown_task_type_is_rejected`
								As expected. 6 automated test(s) passed.	**Pass**

05_IT_01	FR-03	Task that does not match the model type is rejected	Classifier and regressor uploaded.	Choose regression for a classifier and classification for a regressor.	Mismatched pairs	Rejected before evaluation; no metrics shown.	`test_evaluation_service::test_05_IT_03_regression_task_with_a_classifier_is_rejected`
							`test_evaluation_service::test_05_IT_03_classification_task_with_a_regressor_is_rejected`
							`test_api_workflow::test_05_ST_02_mismatched_task_type_is_rejected_over_http`
								As expected. 3 automated test(s) passed.	**Pass**

05_ST_01	FR-03	Task that does not match the dataset target is rejected	Dataset has a continuous or text target.	Choose classification for continuous target; regression for text target.	Mismatched target	Rejected with a reason.	`test_evaluation_service::test_05_ST_02_classification_task_with_a_continuous_target_is_rejected` `test_evaluation_service::test_regression_task_with_a_text_target_is_rejected`	As expected. 2 automated test(s) passed.	**Pass**

UC-06 Evaluate Model

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
06_UT_01	FR-04	Classification metrics match hand calculation	Known model and dataset.	Evaluate; compare each metric with a hand-computed value.	Small fixed dataset	Accuracy, precision, recall, F1 (and ROC-AUC) equal the expected values.	`test_evaluation_service::test_06_UT_03_classification_metrics_match_hand_calculation` `test_evaluation_service::test_classification_result_contains_exactly_the_defined_metrics`	As expected. 2 automated test(s) passed.	**Pass**
06_UT_02	FR-04	Regression metrics match hand calculation	Known model and dataset.	Evaluate; compare MAE, MSE, RMSE, R^2 with hand values.	Small fixed dataset	All four metrics equal the expected values.	`test_evaluation_service::test_06_UT_03_regression_metrics_match_hand_calculation` `test_evaluation_service::test_regression_result_contains_exactly_the_defined_metrics`	As expected. 2 automated test(s) passed.	**Pass**
06_UT_03	FR-04 / FR-05	ROC-AUC skipped with a note when the model has no probabilities; multiclass supported	Model without predict_proba.	Evaluate a classification model that gives no probabilities; evaluate a multiclass one.	No-probability model; 3-class data	ROC-AUC omitted with an explanation; multiclass reports ROC-AUC.	`test_evaluation_service::test_roc_auc_is_skipped_with_a_note_when_model_has_no_probabilities` `test_evaluation_service::test_multiclass_classification_reports_roc_auc`	As expected. 2 automated test(s) passed.	**Pass**
06_IT_01	FR-07	Completed evaluation is stored with its details	Valid model and dataset.	Run an evaluation; read the stored record.	Valid pair	Record holds model, dataset, task, metrics, time and owner.	`test_evaluation_service::test_06_IT_03_completed_evaluation_is_stored_with_its_metadata`	As expected. 1 automated test(s) passed.	**Pass**
06_IT_02	FR-02 / FR-03	Unvalidated or invalid model/dataset cannot be evaluated; nothing is stored	File uploaded but not validated (or invalid).	Try to evaluate.	Unvalidated / invalid files	Rejected with a reason and no evaluation saved.	`test_evaluation_service::test_unvalidated_model_cannot_be_evaluated` `test_evaluation_service::test_unvalidated_dataset_cannot_be_evaluated` `test_evaluation_service::test_model_that_failed_validation_cannot_be_evaluated` `test_api_workflow::test_06_ST_02_evaluating_an_unvalidated_model_is_rejected_and_stores_nothing`	As expected. 4 automated test(s) passed.	**Pass**
06_ST_01	FR-02 / FR-03 / FR-04 / FR-05	End-to-end classification workflow	Logged in as Analyst.	Upload dataset and model, validate, choose classification, evaluate, view results.	Sample classifier and dataset	Metrics and charts are shown.	`test_api_workflow::test_03_ST_01_04_ST_01_06_ST_01_classification_workflow_end_to_end`	As expected. 1 automated test(s) passed.	**Pass**
06_ST_02	FR-04	End-to-end regression workflow; several models on one dataset	Logged in.	Run the regression workflow; evaluate several models on one dataset.	Sample regressor and dataset	Regression metrics shown; each model gets its own evaluation.	`test_api_workflow::test_regression_workflow_end_to_end` `test_api_workflow::test_several_models_can_be_evaluated_against_one_dataset`	As expected. 2 automated test(s) passed.	**Pass**
06_ST_03	NFR-03	Model that crashes while predicting gives a clear error; nothing stored	Model that raises an error on predict.	Evaluate.	Faulty model	Specific error shown; no evaluation saved; app keeps running.	`test_evaluation_service::test_06_ST_02_model_that_crashes_while_predicting_gives_a_clear_error_and_stores_nothing`	As expected. 1 automated test(s) passed.	**Pass**

UC-07 View Metrics

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
07_UT_01	FR-04	Evaluation page shows the right metrics for the task	Completed evaluations exist.	Open a classification and a regression evaluation.	Saved evaluations	Metrics and charts shown; regression page shows only regression metrics.	`test_charts_compare_reports::test_07_UT_01_evaluation_page_shows_metrics_and_charts` `test_charts_compare_reports::test_regression_evaluation_page_shows_only_regression_metrics`	As expected. 2 automated test(s) passed.	**Pass**
07_UT_02	NFR-03	Missing evaluation gives a message, not a crash	Logged in.	Open an evaluation ID that does not exist.	Unknown ID	Clear 'not found' message.	`test_charts_compare_reports::test_07_UT_02_missing_evaluation_gives_a_message_not_a_crash`	As expected. 1 automated test(s) passed.	**Pass**
07_IT_01	FR-07	Reopened evaluation has exactly the stored values	Completed evaluation saved.	Reopen it later.	Saved evaluation	Values equal those first shown.	`test_charts_compare_reports::test_07_IT_01_reopening_gives_exactly_the_stored_values`	As expected. 1 automated test(s) passed.	**Pass**

UC-08 View Visualizations

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
08_UT_01	FR-05	Classification shows confusion matrix and ROC curve	Classification evaluation saved.	Open the evaluation charts.	Classifier with probabilities	Confusion matrix and ROC curve are drawn.	`test_charts_compare_reports::test_08_UT_01_classification_gets_confusion_matrix_and_roc_curve` `test_charts_compare_reports::test_08_UT_02_each_available_chart_renders_a_png`	As expected. 2 automated test(s) passed.	**Pass**
08_UT_02	FR-05	Regression shows residual and predicted-vs-actual plots	Regression evaluation saved.	Open the evaluation charts.	Regressor	Both plots are drawn.	`test_charts_compare_reports::test_08_UT_03_regression_gets_residual_and_predicted_vs_actual_plots`	As expected. 1 automated test(s) passed.	**Pass**
08_IT_01	FR-05	No chart of the wrong task type; unknown chart is 404	Evaluations of both types saved.	Ask for a regression chart on a classification evaluation and vice versa; ask for an unknown chart.	Wrong chart kinds	Not available / 404.	`test_charts_compare_reports::test_08_IT_03_no_classification_chart_for_regression_and_vice_versa` `test_charts_compare_reports::test_unknown_chart_kind_is_404`	As expected. 2 automated test(s) passed.	**Pass**
08_ST_01	FR-05	No ROC curve when the model gives no probabilities	Classifier without predict_proba evaluated.	Open its charts.	No-probability model	Confusion matrix only; no ROC curve.	`test_charts_compare_reports::test_08_ST_02_no_roc_curve_when_the_model_gives_no_probabilities`	As expected. 1 automated test(s) passed.	**Pass**

08_ST_02	FR-05	Chart data agrees with the results	Evaluations saved.	Compare confusion-matrix counts and ROC data with the predictions; check large regression sampling.	Fixed predictions; 50,000-row regression	Counts match; ROC is a valid curve; large charts are sampled but metrics use every row.	`test_charts_compare_reports::test_confusion_matrix_counts_match_the_predictions`
							`test_charts_compare_reports::test_roc_curve_data_is_a_valid_curve`
							`test_charts_compare_reports::test_multiclass_evaluation_gets_one_roc_curve_per_class`
							`test_charts_compare_reports::test_large_regression_charts_use_a_sample_but_metrics_use_every_row`
								As expected. 4 automated test(s) passed.	**Pass**

UC-09 Compare Models

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
09_UT_01	FR-06	Fewer than two evaluations cannot be compared	Evaluations exist.	Compare one evaluation; compare the same one twice; include a missing ID.	1 evaluation / duplicate / missing	Refused with a reason.	`test_charts_compare_reports::test_comparing_fewer_than_two_evaluations_is_refused` `test_charts_compare_reports::test_comparing_the_same_evaluation_twice_counts_as_one` `test_charts_compare_reports::test_comparing_with_a_missing_evaluation_is_refused`	As expected. 3 automated test(s) passed.	**Pass**
09_IT_01	FR-06	Different task types are not compared and the user is warned	One classification and one regression evaluation.	Select both and compare.	Mixed task types	Warning shown; no comparison table.	`test_charts_compare_reports::test_09_IT_01_different_task_types_are_not_compared_and_the_user_is_warned`	As expected. 1 automated test(s) passed.	**Pass**
09_IT_02	FR-06	Same-task evaluations are shown side by side	Two or more same-task evaluations.	Select them and compare.	Two classification evaluations	Metrics aligned side by side; comparison chart renders.	`test_charts_compare_reports::test_09_IT_02_same_task_evaluations_are_compared_side_by_side` `test_charts_compare_reports::test_compare_page_and_chart_render`	As expected. 2 automated test(s) passed.	**Pass**
09_ST_01	FR-06	Comparison never picks a winner	Same-task evaluations compared.	Read the comparison output.	Any compared pair	Values shown with no ranking or 'best' label.	`test_charts_compare_reports::test_comparison_never_ranks_the_models`	As expected. 1 automated test(s) passed.	**Pass**

UC-10 View Evaluation History

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
10_UT_01	FR-07	History is empty for a new user	New account.	Open the history page.	None	Empty-state message shown.	`test_charts_compare_reports::test_10_UT_03_history_starts_empty`	As expected. 1 automated test(s) passed.	**Pass**
10_UT_02	FR-07	History lists newest first	Several evaluations saved.	Open the history page.	3 evaluations	Newest at the top; each links to its saved result.	`test_charts_compare_reports::test_10_UT_02_history_lists_newest_first`	As expected. 1 automated test(s) passed.	**Pass**

UC-11 Generate Report

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
11_UT_01	FR-07	Single-evaluation PDF report	Evaluation saved.	Download the report for one evaluation.	One saved evaluation	Valid PDF with model, dataset, metrics and charts.	`test_charts_compare_reports::test_11_UT_01_single_evaluation_report_is_a_pdf` `test_charts_compare_reports::test_report_embeds_every_chart_as_an_image`	As expected. 2 automated test(s) passed.	**Pass**
11_UT_02	FR-07	Multi-evaluation PDF covers all selected evaluations	Several evaluations saved.	Select several; download one report.	3 evaluations	Report includes every selected evaluation.	`test_charts_compare_reports::test_11_UT_02_multi_evaluation_report_covers_all_of_them`	As expected. 1 automated test(s) passed.	**Pass**
11_UT_03	FR-07	No selection gives a message	Logged in.	Request a report with nothing selected.	Empty selection	Clear message; no file.	`test_charts_compare_reports::test_11_UT_03_no_selection_gives_a_message`	As expected. 1 automated test(s) passed.	**Pass**
11_IT_01	FR-07	Report is built from saved data, not recomputed	Evaluation saved.	Generate the report twice.	Saved evaluation	Same content both times; model is not re-run.	`test_charts_compare_reports::test_11_IT_02_report_is_built_from_saved_data_not_recomputed` `test_charts_compare_reports::test_report_is_the_same_for_the_same_saved_evaluation`	As expected. 2 automated test(s) passed.	**Pass**
11_ST_01	FR-07 / NFR-03	Report refused when an evaluation no longer exists	Evaluation removed.	Request a report including it.	Deleted evaluation ID	Refused with a clear reason.	`test_charts_compare_reports::test_11_ST_02_report_is_refused_when_an_evaluation_no_longer_exists` `test_compatibility_at_upload::test_deleting_a_model_keeps_past_evaluations_readable`	As expected. 2 automated test(s) passed.	**Pass**

UC-12 Manage User Accounts

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
12_UT_01	FR-08	Admin sees all accounts with status and role	Admin logged in; several accounts.	Open the user list.	3 accounts	Every account listed with role and status.	`test_admin::test_12_UT_01_admin_sees_all_accounts_with_status_and_role`	As expected. 1 automated test(s) passed.	**Pass**
12_UT_02	FR-08	Admin can suspend and reactivate an account	Active Analyst account.	Suspend; then reactivate.	Analyst account	Status changes each time and the change is logged.	`test_admin::test_12_UT_02_admin_can_suspend_and_12_UT_03_reactivate`	As expected. 1 automated test(s) passed.	**Pass**
12_IT_01	FR-08	Admin cannot suspend their own account	Admin logged in.	Try to suspend own account.	Own account	Refused; account stays active.	`test_admin::test_12_IT_01_admin_cannot_suspend_their_own_account`	As expected. 1 automated test(s) passed.	**Pass**
12_IT_02	FR-08	Last active Admin cannot be suspended	Only one active Admin (another Admin is suspended).	Try to suspend the last active Admin.	Last Admin	Refused; an Admin always remains.	`test_admin::test_12_IT_02_last_active_admin_cannot_be_suspended` `test_admin::test_suspended_admin_cannot_use_admin_functions`	As expected. 2 automated test(s) passed.	**Pass**

12_ST_01	FR-08 / NFR-03	Invalid status request or unknown account is refused	Admin logged in.	Set a status the account already has; send an invalid status; target an unknown ID.	Same status / 'banned' / bad ID	Refused with a reason (404 for unknown account); nothing logged.	`test_admin::test_12_ST_02_setting_the_state_an_account_already_has_is_refused`
							`test_admin::test_status_must_be_active_or_suspended`
							`test_admin::test_unknown_account_returns_404`
								As expected. 3 automated test(s) passed.	**Pass**

UC-13 Manage User Roles

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
13_UT_01	FR-08	Admin can promote an Analyst to Admin and back	Admin logged in; Analyst exists.	Change role to Admin; change back.	Analyst account	Role changes at once each time.	`test_admin::test_13_UT_03_admin_can_make_an_analyst_an_admin_and_back`	As expected. 1 automated test(s) passed.	**Pass**
13_IT_01	FR-08	Admin cannot change their own role	Admin logged in.	Try to demote own account.	Own account	Refused.	`test_admin::test_13_IT_03_admin_cannot_change_their_own_role`	As expected. 1 automated test(s) passed.	**Pass**
13_IT_02	FR-08	Last active Admin cannot be demoted	Only one active Admin.	Try to demote the last Admin.	Last Admin	Refused.	`test_admin::test_13_ST_02_last_active_admin_cannot_be_demoted`	As expected. 1 automated test(s) passed.	**Pass**
13_IT_03	FR-08	Every change is logged with who, what and when; refused changes are not	Admin logged in.	Change roles and statuses; make a refused change; open the audit log.	Several changes	Log shows actor, target, old/new value and time; refused changes are absent.	`test_admin::test_13_IT_02_every_change_is_logged_with_who_what_and_when` `test_admin::test_refused_changes_are_not_logged` `test_admin::test_audit_log_page_lists_entries`	As expected. 3 automated test(s) passed.	**Pass**
13_ST_01	FR-01 / FR-08	Only an Admin can create an Admin; first Admin comes from the command line	Empty database or Admin logged in.	Run create-admin; create an account as Admin and as Analyst.	create-admin command	First Admin is created; Analysts cannot create accounts; creation is logged.	`test_admin::test_first_admin_can_be_created_from_the_command_line` `test_admin::test_only_an_admin_can_create_an_admin_account` `test_admin::test_account_creation_is_logged`	As expected. 3 automated test(s) passed.	**Pass**

5.3 Non-Functional Test Cases

Performance (NFR-01), reliability (NFR-03), usability (NFR-04) and extensibility (NFR-05).

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
NF_PF_01	NFR-01	Classification on 50,000 rows finishes within 10 s	Single user; model and 50,000-row dataset validated.	Evaluate and time the request.	50,000 rows, typical models	Completes in under 10 s.	`test_performance::test_classification_at_50000_rows_finishes_within_10_seconds`	As expected. 2 automated test(s) passed. Each run, including data set-up, took under 5.6 s (limit 10 s).	**Pass**
NF_PF_02	NFR-01	Regression on 50,000 rows finishes within 10 s	Single user; model and 50,000-row dataset validated.	Evaluate and time the request; open the charts.	50,000 rows	Under 10 s; charts are sampled.	`test_performance::test_regression_at_50000_rows_finishes_within_10_seconds_and_charts_are_sampled`	As expected. 1 automated test(s) passed. Each run, including data set-up, took under 5.6 s (limit 10 s).	**Pass**
NF_PF_03	NFR-01	Evaluation with a model close to 100 MB finishes within 10 s (manual)	A real ~100 MB model and a 50,000-row dataset.	Upload, validate, evaluate; time the request.	~100 MB model	Under 10 s, or the result is recorded as a known limit.	Manual (not automated)	As expected. Run on 4 Nov 2026: model 95.8 MB (1,112 trees), 50,000 rows, 5 evaluations took 1.59, 1.60, 1.62, 1.64, 1.67 s (limit 10 s; 5 of 5 within). Output saved in perf-100mb-output.txt.	**Pass**
NF_ER_01	NFR-03	Malformed or non-object JSON is rejected clearly	Logged in.	Send an evaluation request as non-JSON, broken JSON, and a JSON list.	Bad request bodies	Each rejected with a specific JSON error.	`test_api_workflow::test_evaluation_request_must_be_json` `test_api_workflow::test_malformed_json_body_is_rejected` `test_api_workflow::test_json_body_that_is_not_an_object_is_rejected`	As expected. 3 automated test(s) passed.	**Pass**
NF_ER_02	NFR-03	Missing, blank or unknown evaluation inputs are named in the error	Logged in.	Send an evaluation with a field missing, blank, not text, and with unknown IDs.	Bad evaluation fields	Error names the field; unknown IDs give a clean 400.	`test_api_workflow::test_missing_evaluation_fields_are_named_in_the_error` `test_api_workflow::test_blank_and_non_string_evaluation_fields_are_rejected` `test_api_workflow::test_evaluation_with_unknown_ids_is_a_clean_400`	As expected. 3 automated test(s) passed.	**Pass**
NF_ER_03	NFR-03	Unknown route, wrong method and server errors return safe JSON	App running.	Open an unknown URL; use the wrong HTTP method; force an unexpected error.	Unknown path / wrong method / forced error	404 and 405 as JSON; 500 is generic and leaks no internal details.	`test_api_workflow::test_unknown_route_returns_json_404` `test_api_workflow::test_wrong_http_method_returns_json_405` `test_api_workflow::test_unexpected_server_error_returns_generic_500_without_leaking_details`	As expected. 3 automated test(s) passed.	**Pass**
NF_ER_04	NFR-03	Rejected uploads leave nothing stored	Logged in.	Upload invalid files; then list models and datasets.	Invalid files	Lists contain no entry for the rejected files.	`test_api_workflow::test_nothing_is_stored_for_rejected_uploads`	As expected. 1 automated test(s) passed.	**Pass**

Test Case ID	SRS Req.	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Automated test (backend/tests)	Actual Result	Test Result
NF_US_01	NFR-04	A new user reaches evaluation results unassisted (manual usability check)	New account; sample files; person who has not seen the app.	Ask the person to register, upload, validate, evaluate and open results without help; note where they hesitate.	sample_data folder	Completes the flow unaided. Any step that needs help is recorded as a defect.	Manual (usability check)	As expected. Run on 4 Nov 2026 with one classmate who is not on the team and had not seen the app. Finished register, upload, validate, evaluate and results unaided in about 2-3 minutes with no hesitation. No defects.	**Pass**

NF_US_02	NFR-04	All pages render and expected form fields exist	Logged in.	Open every page; check forms and static files load.	None	Pages render; forms have the fields the script expects.	`test_app::test_workspace_pages_render_for_a_signed_in_analyst`
							`test_app::test_page_forms_have_the_ids_the_script_expects`
							`test_app::test_static_assets_are_served`
							`test_app::test_login_and_register_pages_render_without_an_account`
								As expected. 8 automated test(s) passed.	**Pass**

NF_EX_01	NFR-05	A new chart can be added by changing only the chart module	Source code available.	Register a new chart drawer and title; run the suite.	Test chart	No change needed in upload, storage or login code.	`test_charts_compare_reports::test_adding_a_chart_only_needs_a_drawer_and_a_title`	As expected. 1 automated test(s) passed.	**Pass**

6. Changes from the Earlier Test Plan

Area	What the system now does, and where it is tested
Scope of the plan	The earlier plan had 34 test cases for UC-01 to UC-05, all 'Not Executed'. This plan has 86 cases for UC-01 to UC-13 plus security and non-functional requirements.
Traceability	Every case names its SRS requirement; section 2 gives the reverse view.
Registration and roles (UC-01, UC-13)	Registration creates Analysts only. The first Admin is created from the command line and other Admins only by an Admin (01_*, SEC_02, 13_ST_01).
Data ownership (NFR-02)	Each Analyst sees only their own data; other users' IDs return 404 (SEC_08 to SEC_12).
Compatibility (UC-04)	Also reported when a file is validated, not only at evaluation (04_IT_03).
Results (UC-07 to UC-11)	Confusion matrix, ROC, residual and predicted-vs-actual charts; saved history; comparison without ranking; PDF reports.
Admin (UC-12, UC-13)	Account list, suspend and reactivate, role change, audit log, last-admin protection.
Execution	Automated tests are mapped to each case. Results are recorded in BL-13.

7. Test Execution Summary (BL-13)

Item	Result
Date and time of run	4 November 2026, 20:50 IST
Environment	Windows, Python 3.14.6, pytest 9.1.1, run from the backend folder on the main branch
Command	python -m pytest tests -v -p no:cacheprovider --junitxml=junit.xml
Automated suite	219 tests run, 219 passed, 0 failed, 0 errors, 0 skipped, total time 56.9 s
Test cases in this plan	86: 86 Pass, 0 Fail, 0 not yet executed
Supporting tests	The suite also contains tests that are not listed against a case (for example hashing, e-mail case handling, duplicate-row counting). They also passed.
Performance (NFR-01)	All three 50,000-row runs (logistic regression, random forest, regression) passed the 10 s limit. The manual case with a 95.8 MB model and 50,000 rows (NF_PF_03) also passed: five runs took 1.59 to 1.67 s.
Usability (NFR-04)	One classmate who had not seen the app finished the whole flow unaided in about 2-3 minutes with no hesitation (NF_US_01). One tester only.
Defects found	None in this run.
Evidence	pytest-verbose.txt, junit.xml and perf-100mb-output.txt (docs/test-results)